

Data Analyst Job Postings: Experience, Salary, and Data Languages

Introduction

This report explores relationships between minimum years of experience, median salary estimates, and data language requirements in data analyst job postings. The analysis includes a scatter plot showing the relationship between experience and salary, and a box plot comparing salary distributions across data language categories (R, Python, Both, and Neither). Because the dataset is relatively small, findings should be interpreted cautiously.

Visualization and Interpretation



Figure 1. Scatter plot showing the relationship between minimum years of experience required and median salary estimate.

Salaries tend to increase with more experience, although there is noticeable variability at different experience levels. Some lower-experience positions still report relatively high salary estimates. The limited sample size means that a few outliers may strongly influence the observed pattern.

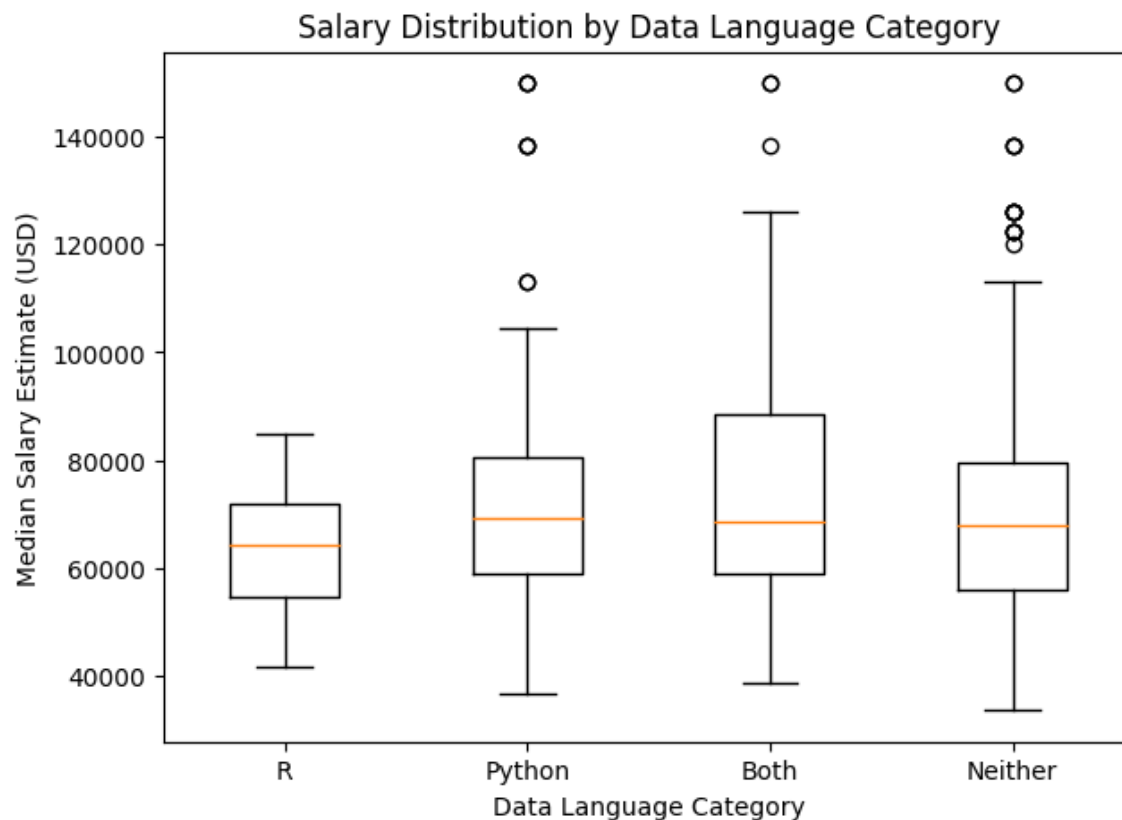


Figure 2. Box plot comparing median salary estimates across data language categories: R, Python, Both, and Neither.

Job postings requiring both R and Python appear to show higher typical salary estimates compared to some other categories. However, salary ranges overlap substantially across groups, and the small sample size limits the reliability of strong conclusions.

Limitations

This analysis is based on a relatively small sample of job postings. As a result, findings may not fully represent the broader job market, and individual outliers can significantly affect observed trends.

My reflection

This project helped me better understand how AI tools and APIs can support data analysis and visualization tasks. One of the easier parts was generating graphs and cleaning the dataset once the correct prompts were provided. However, it was sometimes difficult to make sure the data was formatted correctly, especially when dealing with missing or non-

numeric values in the experience column. I also had to adjust the plot titles and labels to make the visuals clearer and more professional.

What surprised me most was how quickly OpenRouter AI and APIs could assist with coding, data cleaning, and generating interpretations. I learned that AI can save time during analysis, but it still requires careful review to ensure the results are accurate and meaningful.